

Mathematical Induction (Divisibility) - Development Questions

- 1 Use the Principle of Mathematical Induction to verify that, for any positive integer n , 6^{n-1} is divisible by 5. 3
- 2 Use the Principle of Mathematical Induction to prove that $4^n + 5^n + 6^n$ is divisible by 15 for all **ODD** positive integers n . 3
- 3 Use the Principle of Mathematical Induction to prove that $n^3 - n + 3$ is divisible by 3 for all positive integers n . 3
- 4 Use the Principle of Mathematical Induction to prove that 2^{2n-1} is divisible by 3 for all positive integers n . 3
- 5 Use the Principle of Mathematical Induction to prove that $9^n + 3$ is divisible by 4 for all positive integers n . 3
- 6 Use the Principle of Mathematical Induction to prove that $8^n - 3^n$ is divisible by 5 for all positive integers n . 3
- 7 Use the Principle of Mathematical Induction to prove that $5^{2n-1} + 1$ is divisible by 6 for all positive integers n . 3
- 8 Use the Principle of Mathematical Induction to prove that $5^n + 8n + 3$ is divisible by 4 for all positive integers n . 3
- 9 Use the Principle of Mathematical Induction to prove that $8^n - 2^n$ is divisible by 6 for all positive integers n . 3

10 Prove, using Mathematical Induction, that the sum of the cubes of three consecutive integers is divisible by 3 **3**

11 Prove that $n^2 + n$ is divisible by 2 for all positive integers n using Mathematical Induction. **3**

12 Use the Principle of Mathematical Induction to prove that $4^n - 1$ is divisible by 3 for all $n \geq 1$. **3**

13 Prove that $3^{3n} + 2^{n+2}$ is divisible by 5 for all $n \geq 1$ using Mathematical Induction. **3**

14 Consider the function below: **3**

$$f(n) = (4n + 3)5^n - 3$$

Prove by induction that $f(n)$ is divisible by 16, for all integers $n \geq 1$

15 Prove $15^n - 8^{n-2}$ is divisible by 7 for integers $n \geq 2$, **3**

16 Prove $4^n + 6n - 1$ is divisible by 9 for all positive integers n **3**

17 Prove by induction that for all positive integers n , $4^{n+1} + 5^{2n-1}$ is divisible by 21 **3**

18 Prove by mathematical induction that if n is a positive integer, then, **3**

$$3^{2n+3} + 2^{n+3}$$

is divisible by 8.

19 Consider the function $f(n)$ below:

$$f(n) = n^3 + 5n$$

(a) Show that $n^2 + n + 2$ is always even for n is a positive integer **1**

(b) Hence, prove by Mathematical Induction that $f(n)$ is divisible by 6 for the positive integers $n \geq 1$. **3**

20 Prove by mathematical induction that $5^{2n^2} - 1$ is divisible by 4. **3**